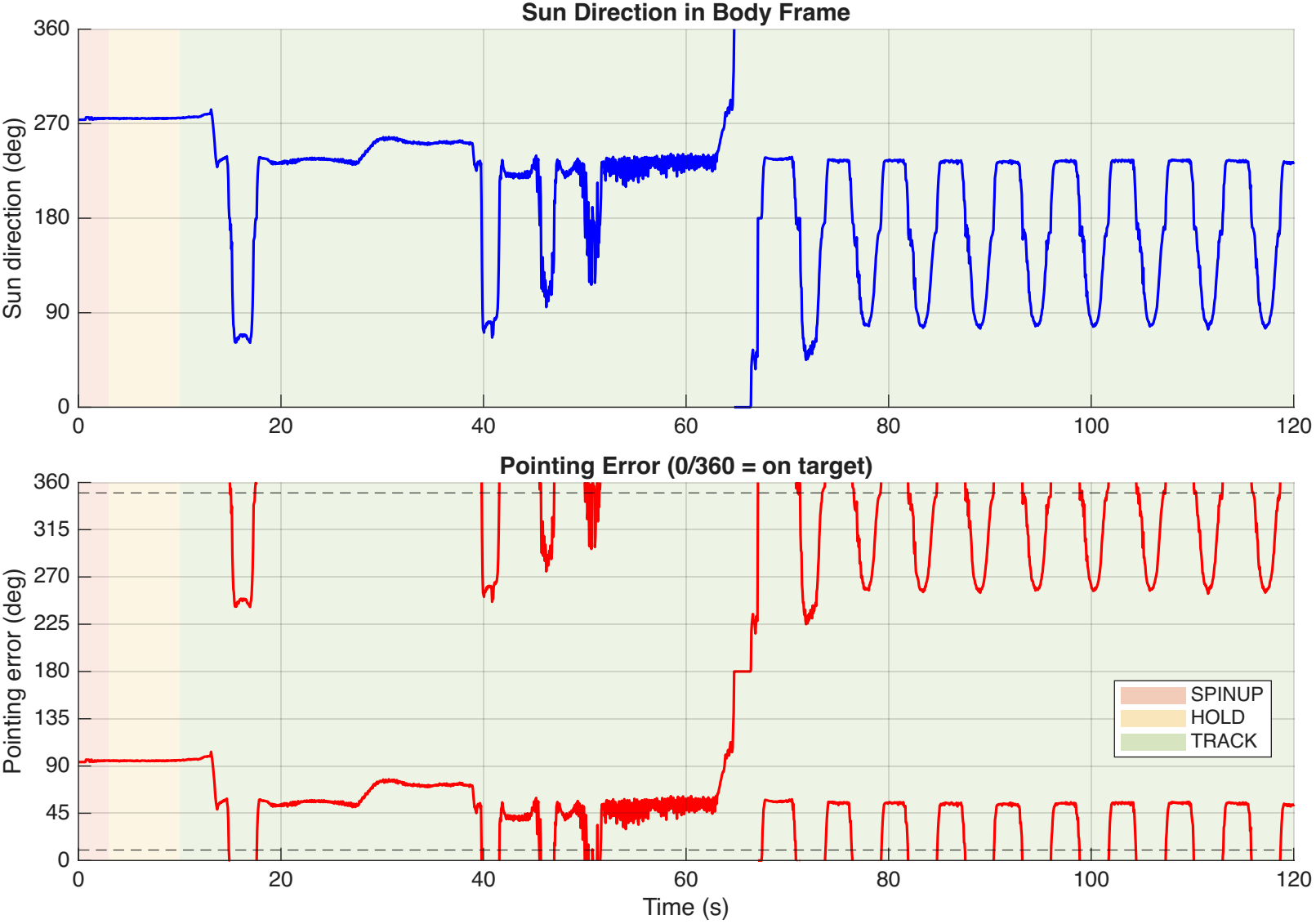
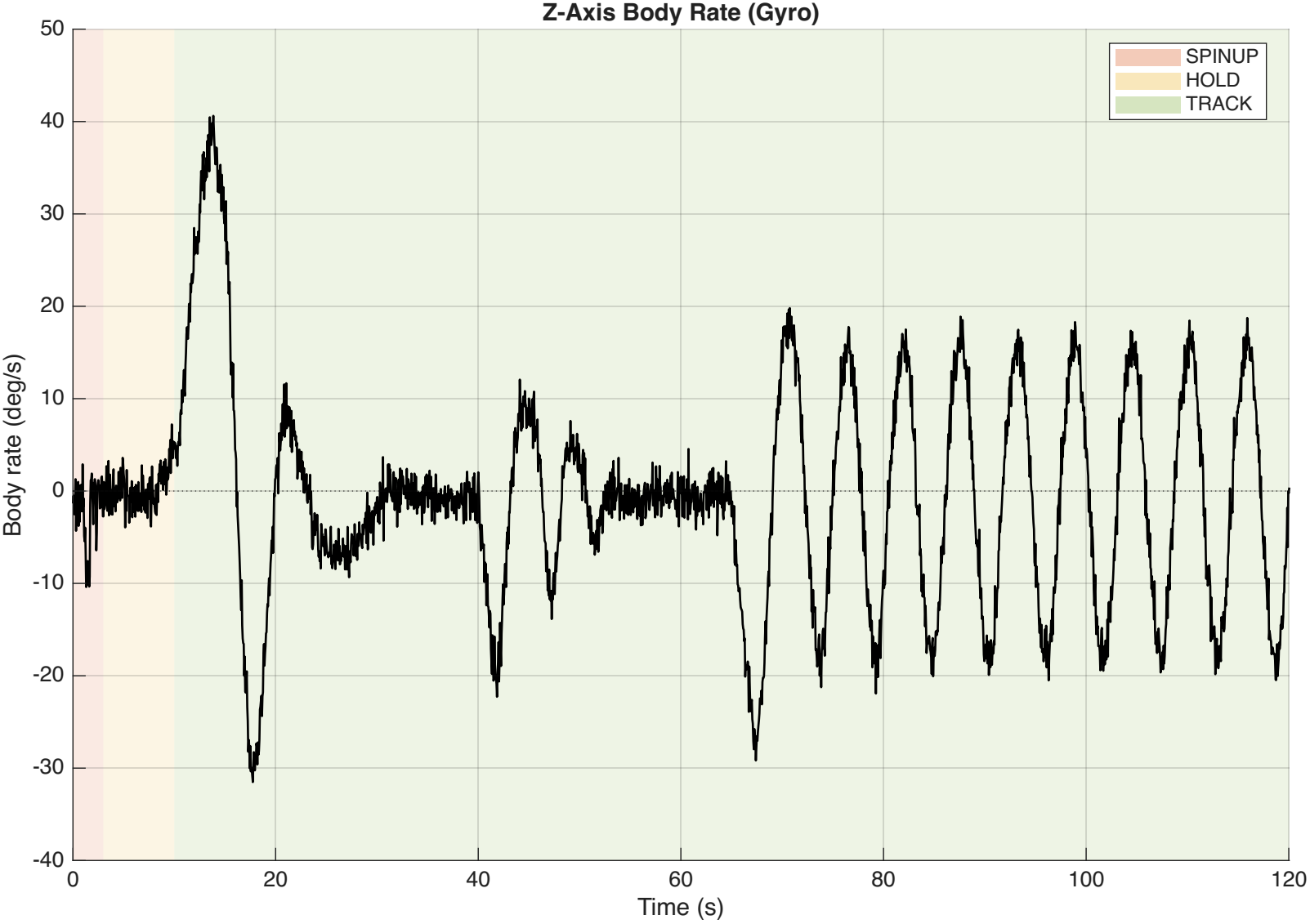
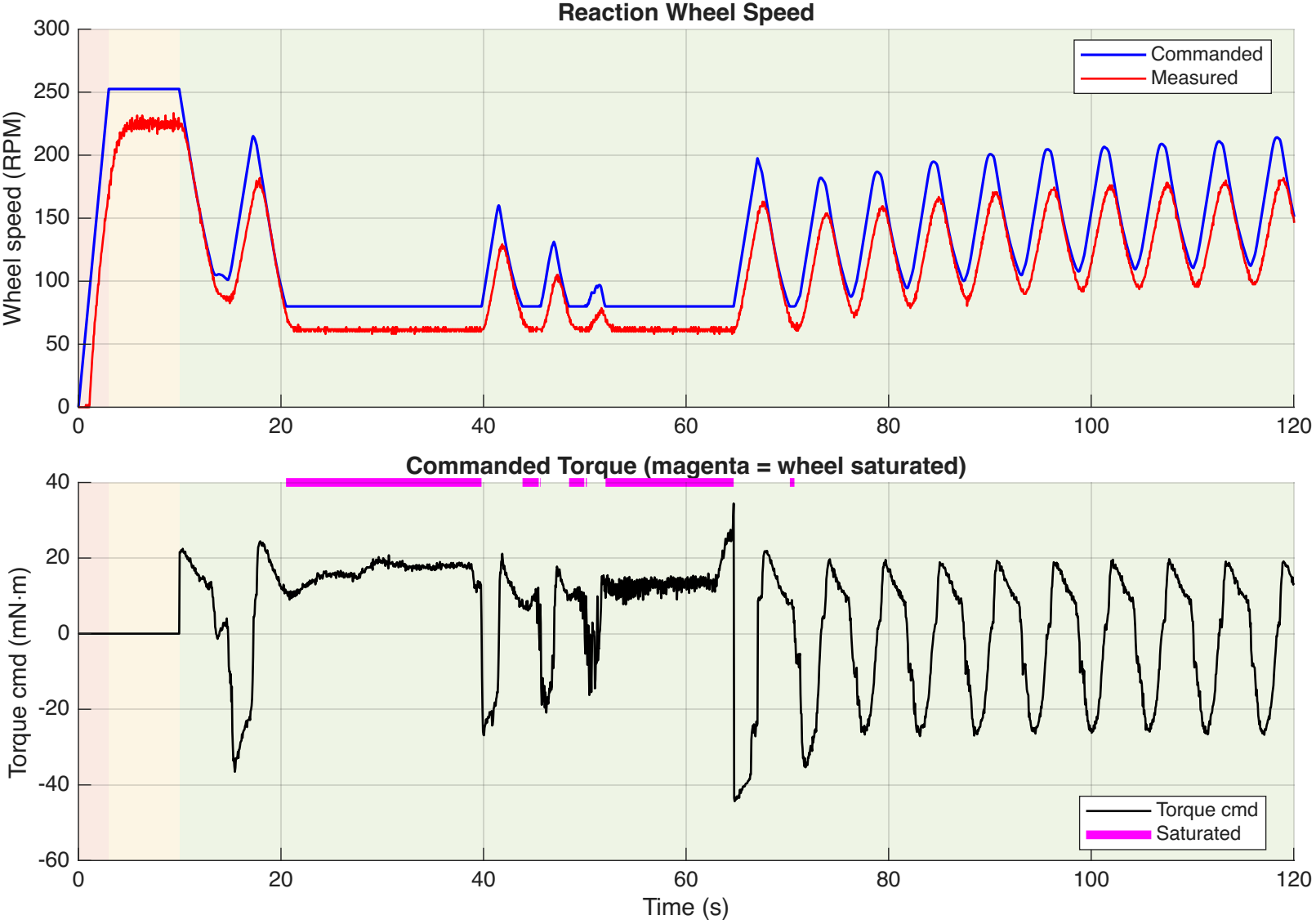
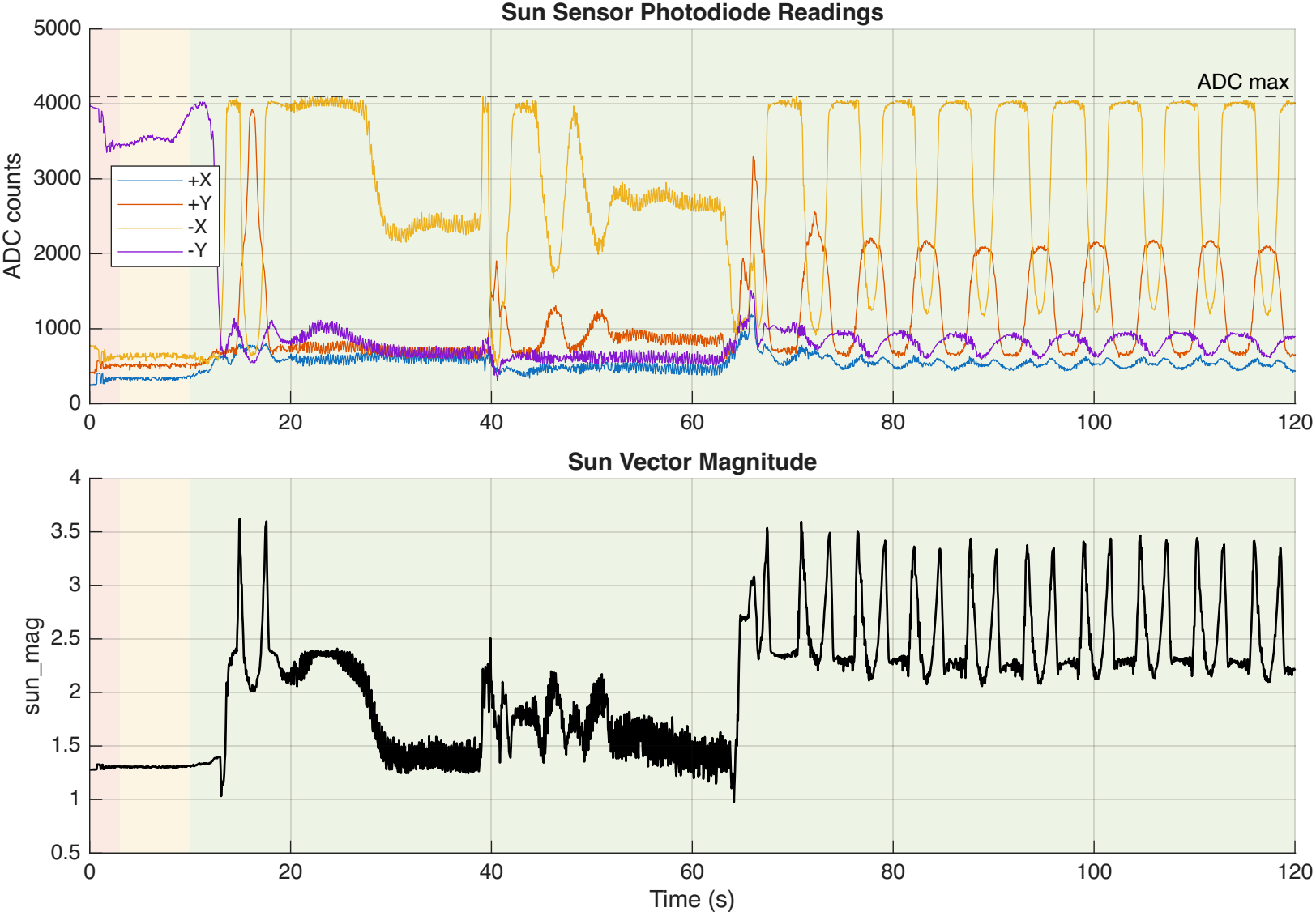
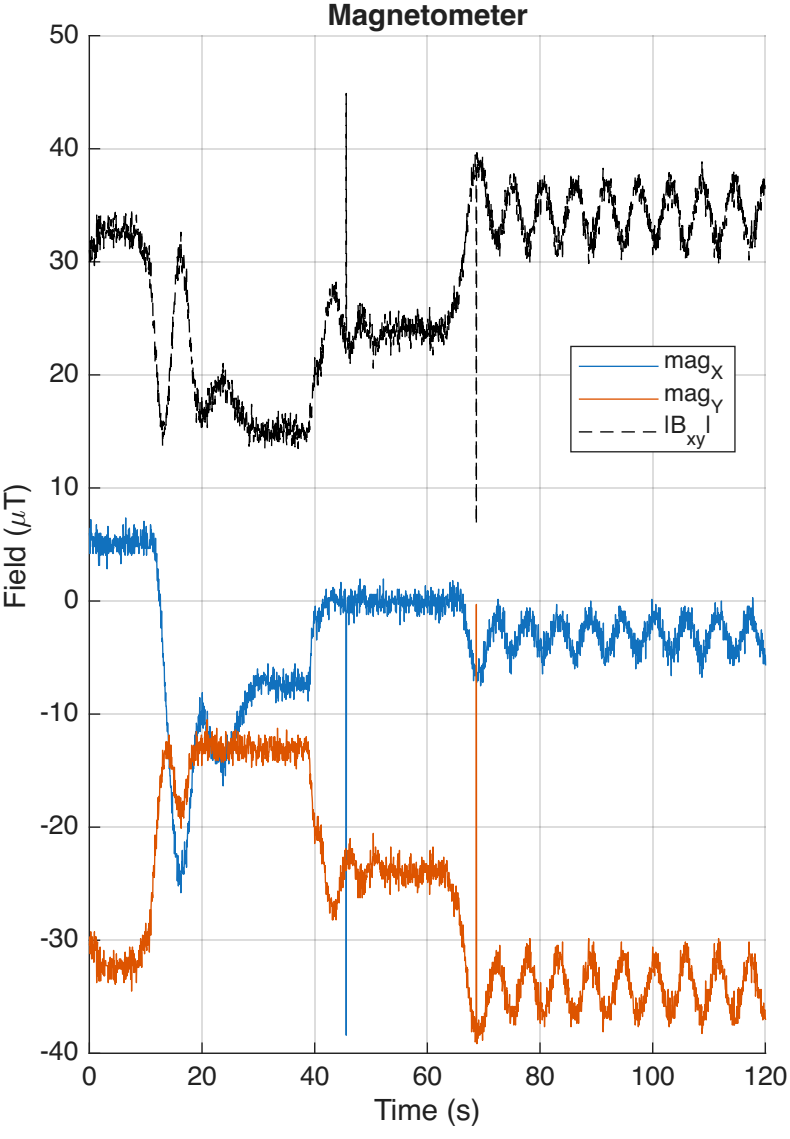
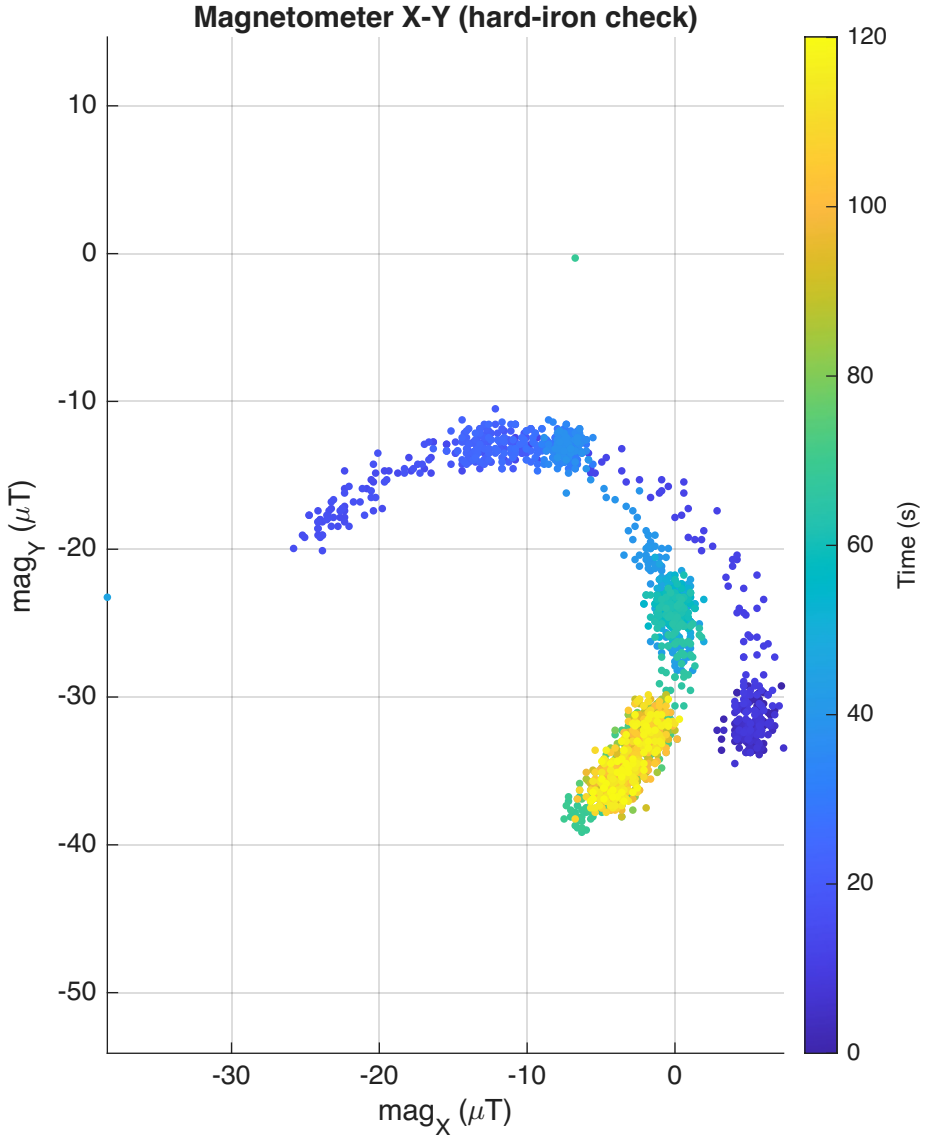

Sun Direction in Body Frame
Sun direction (deg)
360
270
180
90
0
0
20
40
60
80
100
120
Pointing Error (0/360 = on target)
Pointing error (deg)
360
315
270
225
180
135
90
45
0
SPINUP
HOLD
TRACK
Time (s)

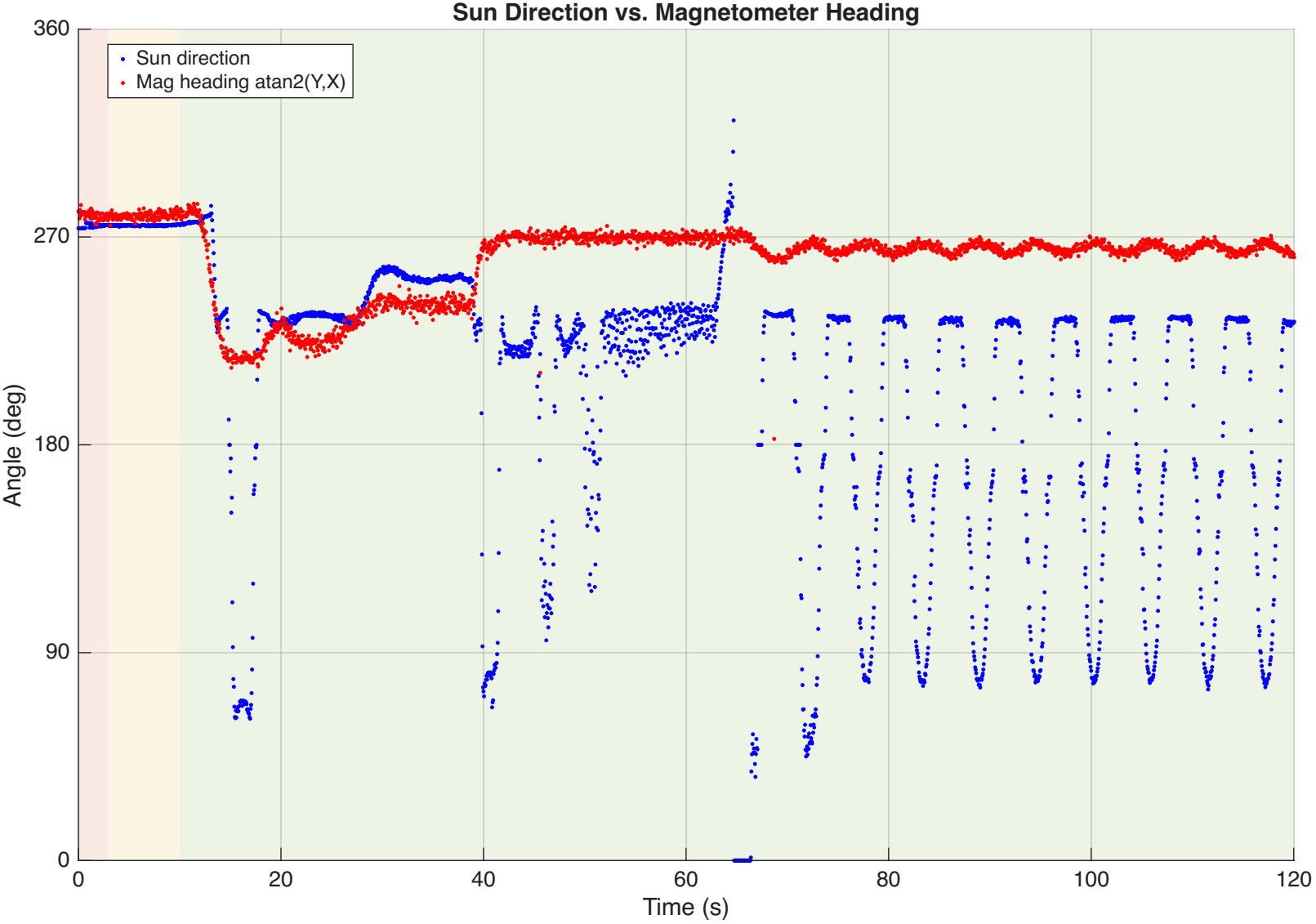

Z-Axis Body Rate (Gyro)
SPINUP
HOLD
TRACK
Body rate (deg/s)
Time (s)
50
40
30
20
10
0
-10
-20
-30
-40
0
20
40
60
80
100
120

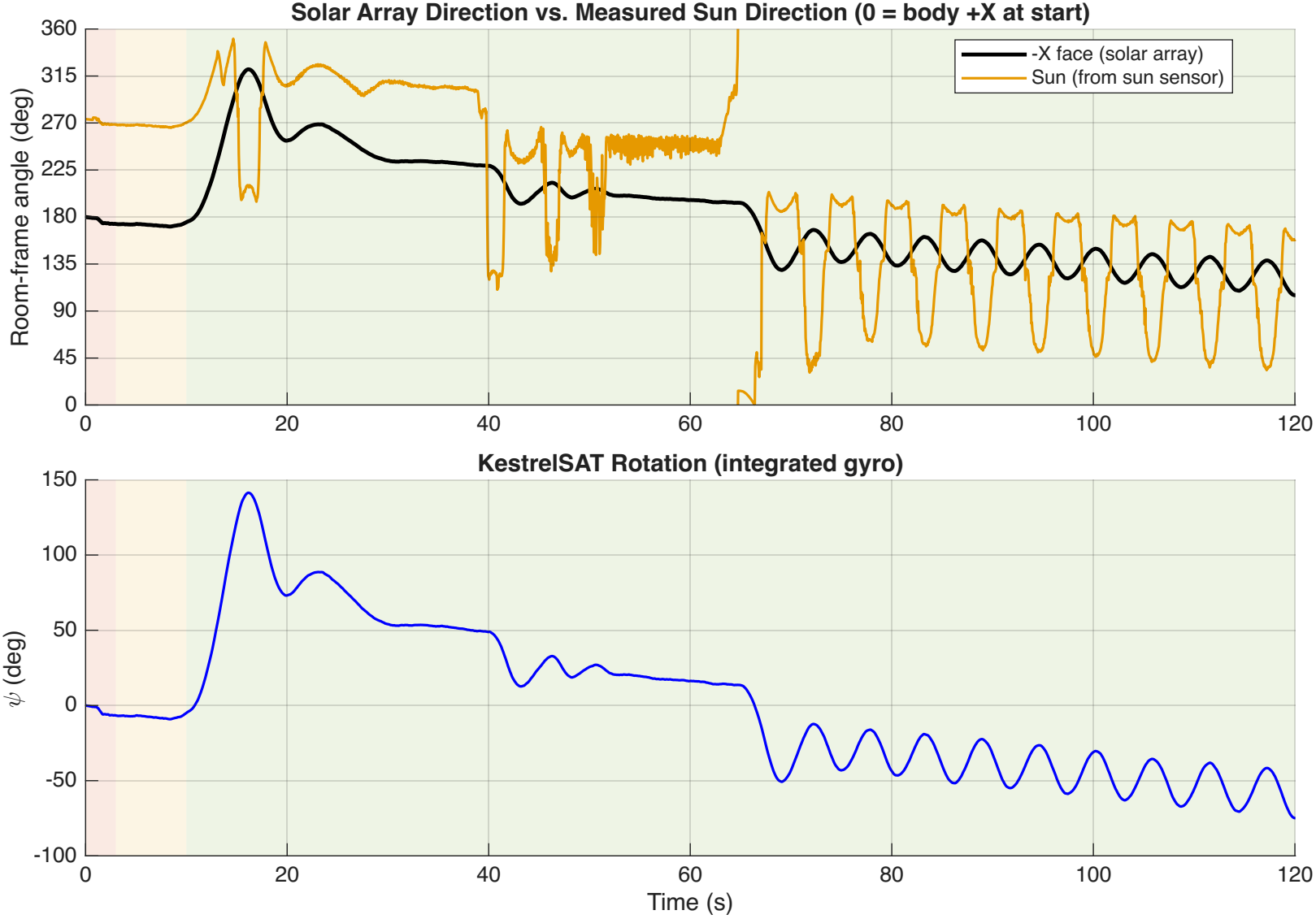

Reaction Wheel Speed
Commanded
Measured
Wheel speed (RPM)
300
250
200
150
100
50
0
0
20
40
60
80
100
120
Commanded Torque (magenta = wheel saturated)
Torque cmd (mN·m)
40
20
0
-20
-40
-60
Torque cmd
Saturated
Time (s)


Sun Sensor Photodiode Readings
ADC counts
ADC max
+X
+Y
-X
-Y
5000
4000
3000
2000
1000
0
0
20
40
60
80
100
120
Sun Vector Magnitude
sun_mag
4
3.5
3
2.5
2
1.5
1
0.5
Time (s)


Magnetometer
Field (μT)
Time (s)
magX
magY
|Bxy|


Magnetometer X-Y (hard-iron check)
magY (μT)
magX (μT)
Time (s)


Sun Direction vs. Magnetometer Heading
Sun direction
Mag heading atan2(Y,X)
Angle (deg)
Time (s)
360
270
180
90
0
0
20
40
60
80
100
120


Solar Array Direction vs. Measured Sun Direction (0 = body +X at start)
-X face (solar array)
Sun (from sun sensor)
Room-frame angle (deg)
360
315
270
225
180
135
90
45
0
0
20
40
60
80
100
120
KestrelSAT Rotation (integrated gyro)
ψ (deg)
150
100
50
0
-50
-100
0
20
40
60
80
100
120
Time (s)